

Partho Bala

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Summary

Senior AI Backend Engineer with 4+ years of experience building production-grade ML infrastructure, agentic RAG architectures, and scalable API services. Expertise spans the full AI development lifecycle—from training DLRM recommendation systems and fine-tuning transformer models to deploying high-throughput FastAPI microservices and Redis-backed pipelines in Docker environments. Specialized in vector search, LLM orchestration, and robust system design.

Experience

Bangopure Limited

April 2026 – Present

Senior AI Engineer

UK (Remote)

- Designed and developed an AI-powered **Digital Product Passport (DPP)** evidence verification platform to validate manufacturer sustainability and compliance claims.
- Researched and integrated **LLMs**, OCR, and document intelligence techniques to automate evidence extraction and confidence-based claim verification.

Technonext Software Ltd.

Jun 2022 – Nov 2025

AI Engineer

Dhaka, Bangladesh

- Foodi Recommendation System:** Trained and optimized DLRM recommendation models using NVIDIA Merlin, performed feature engineering, and resolved ranking pipeline issues to improve recommendation score accuracy.
- US-Bangla Airlines Travel Assistant Chatbot:** Built an ASR-powered chatbot for a production mobile application by collecting and preprocessing conversational data for Rasa, supporting flight search and baggage-related queries.
- Cartup Search Platform:** Improved product discovery by implementing semantic search with LLMs and embedding-based retrieval, and developed CLIP-based visual search for similar product retrieval.
- Handwritten Document Extraction:** Designed and deployed an OCR pipeline for handwritten English documents, achieving 86%+ extraction accuracy across 5+ invoice formats.
- CarryBee Address Extraction:** Developed a BERT-based Named Entity Recognition (NER) system to extract structured location information, including city and area, from unstructured delivery addresses.
- Developed scalable ML inference APIs using FastAPI and Docker for production model serving.

Swosti Limited

Jan 2022 – May 2022

Junior Software Engineer

Dhaka, Bangladesh

- Maintained and improved enterprise applications by implementing critical bug fixes and feature enhancements.
- Designed and developed business reports using BIRT and Jasper Reports for internal analytics and operational workflows.

Education

Gopalganj Science and Technology University

Jan 2016– Dec 2019

Bsc in Computer Science and Engineering

Gopalganj, Bangladesh

Publication

Analyzing Employee Retention Factors Using Machine Learning Moshir Rahman, Md Rashedul Islam, **Partho Bala**, Abdus Sattar — IEEE ICAECT 2024 DOI: 10.1109/ICAECT60202.2024.10469009

Technical Skills

Programming Language: Python

Machine Learning: PyTorch, TensorFlow, Rasa, Scikit-learn, HuggingFace, NVIDIA Merlin, LangChain, SpaCy Keras, Pandas, Numpy, AWS Lex, Elasticsearch

Backend: FastAPI

Databases: PostgreSQL, Redis

Automation Tools: Selenium, BeautifulSoup

Other: Docker, Git, Jira

Personal Projects

Bengali Clickbait Detection Using NLP | *Python, Scikit-learn, Keras, Gensim*

- Developed an NLP-based Bengali clickbait detection system using traditional machine learning and deep learning approaches.
- Built a custom dataset of clickbait and authentic Bengali news headlines and performed extensive preprocessing and feature engineering.
- Experimented with multiple classification architectures and achieved **81%** classification accuracy.

Country Classifier | *PyTorch, HuggingFace, Pandas, Gradio*

 [GitHub](#)

- Developed a BERT-based text classification system capable of predicting countries from contextual free-form text input.
- Built an interactive inference interface using Gradio for real-time predictions and model demonstration.

Question Answering Bot | *LangChain, FAISS, and Azure OpenAI*

 [GitHub](#)

- Agentic Reasoning: Uses a "Plan-Act-Reflect" cycle to evaluate the relevance of retrieved documents before generating a response.

Distance Learning Platform | *Django, Redis, WebSocket, PostgreSQL*

 [GitHub](#)

- Built an asynchronous e-learning platform supporting course enrollment, real-time group chat, and instructor content management.
- Implemented WebSocket-based communication using Django Channels and Redis channel layers for scalable real-time messaging.
- Developed RESTful APIs with JSON serialization for frontend-backend communication.

Extracurricular Activities

NASA Space Apps Challenge 2019 – Regional Finalist

National High School Programming Contest (NHSPC) – Academic Member and Volunteer (2016–2017)

UGB ICT Carnival – Programming Contest Participant